

Rahul Babu

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🌐 Rahul Babu 📄 Rahul Babu

👤 Profile

Computer Science engineer specializing in backend development and machine learning with expertise in building production-grade conversational AI systems. Proven experience developing scalable multi-tenant SaaS platforms, implementing LLM-powered chatbots, and deploying deep learning models for computer vision applications. Combining strong Python backend development skills with advanced ML knowledge to deliver end-to-end AI solutions.

⚙️ Technical Skills

- **Backend Development:** FastAPI, REST APIs, Multi-tenant SaaS, Microservices, Authentication & RBAC
- **Programming Languages:** Python, JavaScript
- **Machine Learning & AI:** TensorFlow, CNN, LSTM, Transformers, TransUNet, LangChain, LLMs, Computer Vision
- **Databases:** MariaDB, MySQL, MongoDB
- **DevOps & Tools:** Docker, Git, Linux, Raspberry Pi
- **Web Technologies:** HTML, CSS, JavaScript, Vue (basics)

📁 Internship Experience

Backend & Machine Learning Intern
Kochi Digital, Kochi

July 2024 – Present

- Architecting and developing backend services for enterprise-grade SaaS conversational AI platform serving multiple tenants with complete data isolation
- Implemented secure tenant authentication system and role-based access control (RBAC) using FastAPI with JWT tokens
- Designed and deployed 15+ REST API endpoints for user management, analytics dashboards, and chatbot interaction pipelines
- Integrated Large Language Models through LangChain to build domain-specific chatbots with custom knowledge bases and retrieval-augmented generation (RAG)

🔗 Key Projects

Multi-Tenant Conversational AI System (Internship Project): Architected end-to-end SaaS platform enabling organizations to deploy isolated chatbots with configurable knowledge bases, analytics tracking, and custom branding. Implemented secure FastAPI backend with tenant isolation, LangChain-based LLM integration, and React dashboard for conversation analytics and user management.

Technologies: FastAPI, Python, LangChain, MariaDB, React, LLMs, Docker

Liver Tumor Segmentation using TransUNet: Developed automated liver tumor segmentation system using TransUNet architecture combining CNN feature extraction with Transformer encoder for global context. Preprocessed CT scans with HU normalization, implemented patch-based training

pipeline, and achieved 85%+ Dice coefficient for tumor boundary detection.

Technologies: Python, TensorFlow, TransUNet, Medical Image Processing, CT Analysis

Education

Master of Technology in Computer Science and Engineering

CGPA: 9.3/10

St Joseph's College of Engineering and Technology, Palai

2024–2026

Bachelor of Technology in Computer Science and Engineering

CGPA: 7.88/10

St Joseph's College of Engineering and Technology, Palai

2020–2024

Certifications & Professional Development

- Artificial Intelligence and Machine Learning Fundamentals
- English Core Skills Certification – British Council (B2 Level - Upper Intermediate)
- Completed production deployment of multi-tenant SaaS application
- Active contributor to ML and backend development communities